Day 3

Combinational - Assignment

1) Implement the following functions using i) 2:1 MUX and NOT gates (ii) 4:1 MUX.

a) $F(A,B,C) = \bar{A}C + B\bar{C}$

$\Rightarrow$ i>

$= \bar{A}C(B+\bar{B}) + (A+\bar{A})B\bar{C}$

$= \bar{A}BC + \bar{A}\bar{B}C + AB\bar{C} + \bar{A}B\bar{C}$

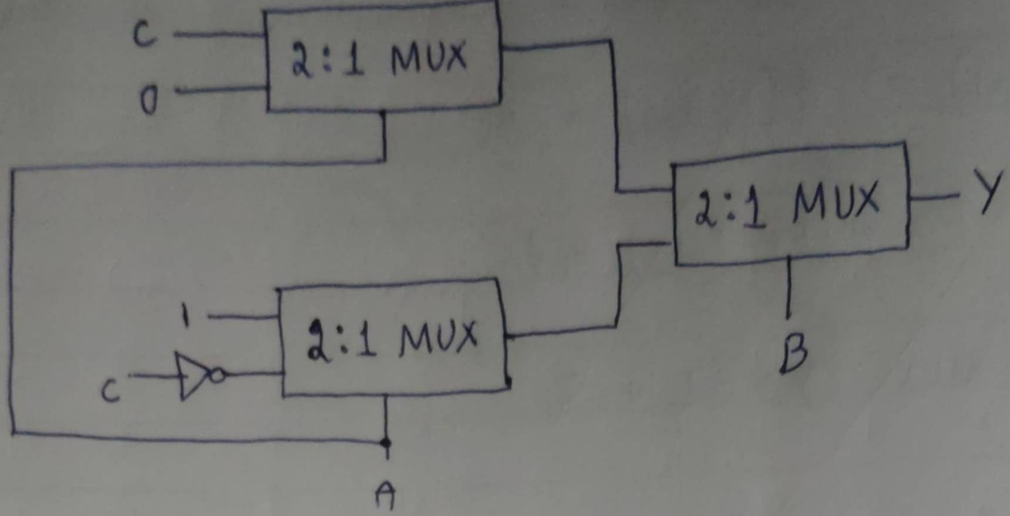
$= m(1,2,3,6)$

Truth Table (using 2:1 MUX and NOT gates):

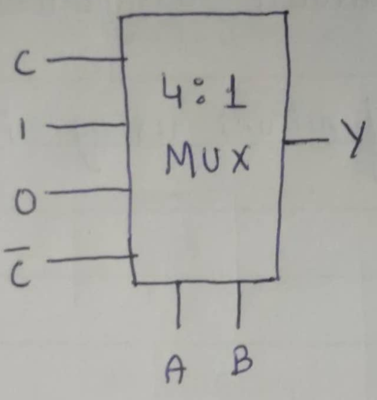
A	B	C	Y (output)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$\Rightarrow$

A	B	Y
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$



ii) Using 4:1 MUX (follow the above truth table)



b) $F(P,Q,R) = \bar{P}\bar{R} + Q\bar{R} + \bar{P}Q$

=> i>

$$= \bar{P}\bar{R}(Q+\bar{Q}) + (P+\bar{P})Q\bar{R} + \bar{P}Q(R+\bar{R})$$

$$= \bar{P}Q\bar{R} + \bar{P}\bar{Q}\bar{R} + PQ\bar{R} + \bar{P}Q\bar{R} + \bar{P}QR + \bar{P}Q\bar{R}$$

$$= \bar{P}Q\bar{R} + \bar{P}\bar{Q}\bar{R} + PQ\bar{R} + \bar{P}QR$$

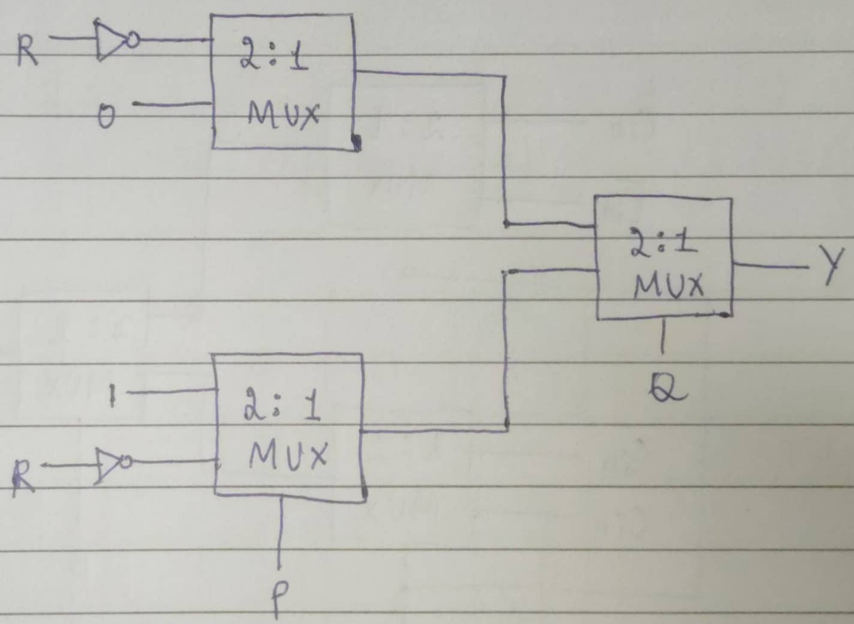
$Y = m(0,2,3,6)$

Truth table (using 2:1 MUX and NOT gate) :

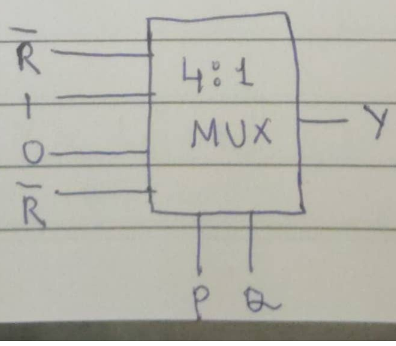
P	Q	R	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

=>

P	Q	Y
0	0	$\bar{R}$
0	1	1
1	0	0
1	1	$\bar{R}$



ii> Using 4:1 Mux :



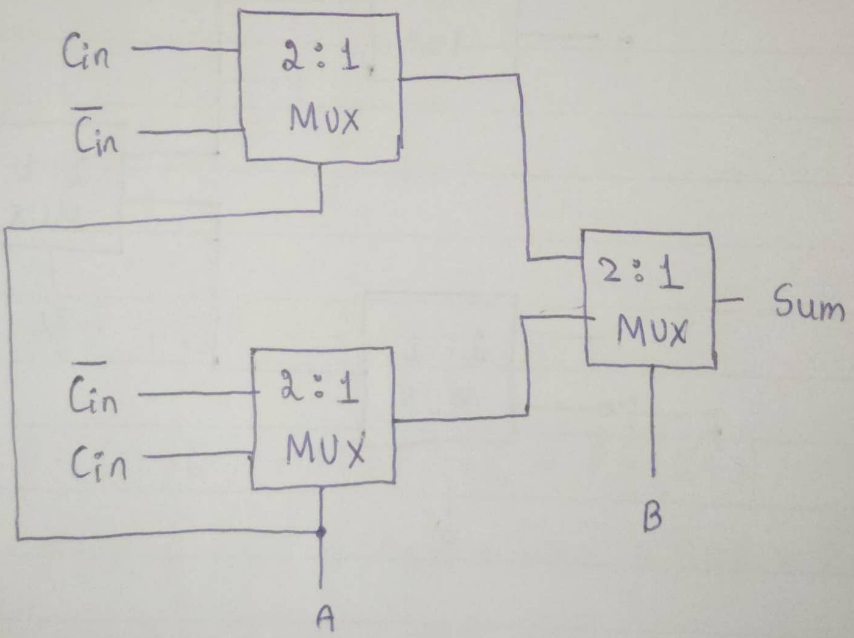
2) Implement the functionality of a full adder using 2:1 Muxes only.

=> Truth table of full_Adder:

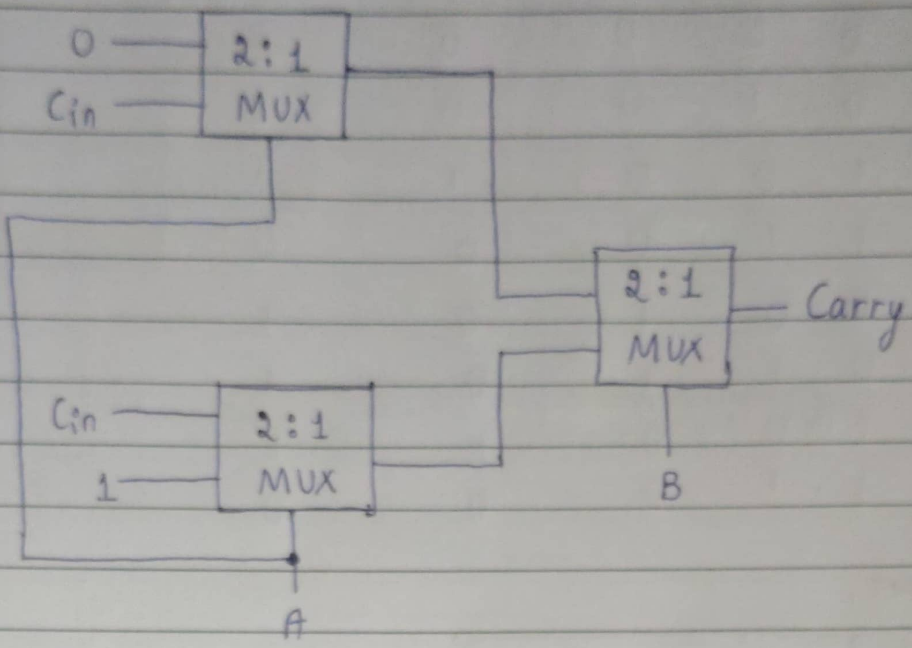
A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

=>

A	B	Sum	Carry
0	0	Cin	0
0	1	$\overline{Cin}$	Cin
1	0	$\overline{Cin}$	Cin
1	1	Cin	1



Expt. No. .... Date



3) Simplify $f(a,b,c,d) = \sum m(2,4,5,6,10) + D(12,13,14,15)$ and implement the SOP expression using 4:1 MUX.

⇒

ab \ cd	$\bar{c}\bar{d}$	$\bar{c}d$	cd	$c\bar{d}$
$\bar{a}\bar{b}$	0 (0)	0 (1)	0 (3)	1 (2)
$\bar{a}b$	1 (4)	1 (5)	0 (7)	1 (6)
ab	X (12)	X (13)	X (15)	X (14)
$a\bar{b}$	0 (8)	0 (9)	0 (11)	1 (10)

$Y = b\bar{c} + c\bar{d}$

Let "b" and "c" be select lines,

b	c	d	$b\bar{c}$	$c\bar{d}$	$b\bar{c}+c\bar{d}$
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

Therefore,

b	c	$b\bar{c}+c\bar{d}=Y$
0	0	0
0	1	$\bar{d}$
1	0	1
1	1	$\bar{d}$